

PYALITSA, Russian Federation

WMO: 223490

Lat: 66.189N

Lon: 39.524E

Elev: 9

StdP: 101.22

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.2	-20.7	-25.7	0.4	-22.9	-23.1	0.5	-20.5	16.1	-6.7	14.7	-6.1	3.0	330	0.872

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.6	18.3	15.0	16.4	13.9	14.8	13.0	15.6	17.4	14.3	15.8	13.2	14.5	5.0	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.6	10.4	16.8	13.4	9.6	15.2	12.5	9.0	14.0	43.7	17.4	40.1	15.9	37.3	14.6	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	12.4	11.1	DB	-25.9	21.6	3.2	2.2	-28.2	23.3	-30.1	24.6	-31.9	25.8	-34.2	27.4	(4)
(5)			WB	-26.4	17.9	3.4	1.8	-28.8	19.1	-30.8	20.2	-32.7	21.2	-35.2	22.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.4	-9.2	-9.4	-7.3	-2.5	2.7	7.5	11.0	10.1	7.3	2.3	-2.6	-5.3
	DBStd	8.17	5.49	6.07	4.76	3.51	3.50	3.22	3.05	2.38	2.57	3.20	3.87	4.72
	HDD10.0	3597	594	543	537	375	228	89	23	28	87	240	378	473
	HDD18.3	6532	853	776	796	625	485	326	227	256	332	498	628	732
	CDD10.0	107	0	0	0	0	2	14	56	31	5	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.2	6.0	5.6	5.2	4.8	4.9	4.7	4.3	4.1	4.7	5.5	6.1	6.4
Precipitation	PrecAvg	450	25	21	20	18	37	39	52	59	51	58	38	32
	PrecMax	637	44	44	36	34	71	99	128	132	107	123	95	62
	PrecMin	324	8	8	5	5	12	10	9	19	13	20	8	14
	PrecStd	69	9	9	9	9	17	20	29	33	25	28	19	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	1.8	2.5	6.3	15.2	18.6	21.9	18.9	14.7	9.2	5.5	2.6
		MCWB	0.0	1.0	0.8	4.0	10.5	14.2	17.5	15.6	11.9	8.3	4.8	1.5
	2%	DB	0.0	0.2	0.7	4.1	11.3	16.0	19.4	16.2	12.9	8.3	4.0	1.8
		MCWB	-0.3	-0.3	-0.2	2.2	8.2	12.4	15.9	14.2	11.2	7.5	3.4	1.1
	5%	DB	-0.6	-0.7	0.0	2.8	9.4	14.1	17.5	14.8	11.9	7.4	3.0	1.3
		MCWB	-1.0	-1.2	-0.5	1.3	6.9	11.5	14.8	13.4	10.9	6.5	2.4	0.7
	10%	DB	-1.7	-1.8	-0.7	1.7	7.8	12.5	15.8	13.6	11.1	6.5	2.2	0.6
		MCWB	-2.1	-2.3	-1.2	0.6	5.9	10.3	13.8	12.4	10.2	5.7	1.6	0.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	1.1	1.0	4.0	11.4	14.8	18.2	16.3	12.9	8.8	5.0	1.9
		MCDB	0.4	1.6	2.0	5.9	14.9	16.9	21.0	17.9	14.2	9.0	5.5	2.2
	2%	WB	-0.3	-0.1	0.1	2.4	8.5	12.9	16.4	14.6	11.7	7.7	3.6	1.3
		MCDB	-0.1	0.0	0.5	3.7	10.8	15.0	18.6	15.9	12.3	8.1	3.9	1.7
	5%	WB	-1.0	-1.1	-0.5	1.5	7.2	11.8	15.2	13.5	11.0	6.6	2.5	0.8
		MCDB	-0.7	-0.7	-0.1	2.4	8.9	14.0	17.0	14.6	11.7	7.2	2.9	1.2
	10%	WB	-2.1	-2.2	-1.2	0.8	6.1	10.5	14.0	12.7	10.4	5.8	1.6	0.1
		MCDB	-1.6	-1.8	-0.7	1.5	7.8	12.3	15.5	13.5	11.1	6.5	2.1	0.5
Mean Daily Temperature Range	5% DB	MDBR	5.0	5.4	6.2	5.4	5.2	5.7	5.6	5.1	4.9	3.9	3.8	4.3
		MCDBR	3.9	4.7	4.7	5.5	8.1	8.3	8.0	6.6	5.6	3.8	3.3	2.9
		MCWBR	3.9	4.6	4.4	4.3	5.5	5.8	5.5	4.8	4.4	3.4	3.4	3.0
	5% WB	MCDBR	3.9	4.6	4.5	4.9	7.4	7.5	7.3	6.0	5.2	3.4	3.4	2.9
		MCWBR	3.9	4.7	4.2	4.0	5.5	5.6	5.4	4.6	4.3	3.3	3.5	3.0
Clear-Sky Solar Irradiance	taub	0.222	0.275	0.306	0.335	0.347	0.342	0.374	0.350	0.328	0.299	0.357	0.366	
	taud	1.938	2.054	2.109	2.181	2.398	2.499	2.434	2.476	2.507	2.299	2.561	2.565	
	Ebn at Noon	191	556	738	819	855	870	823	809	743	547	202	47	
	Edh at Noon	18	58	90	109	99	92	95	83	64	46	18	8	
All-Sky Solar Radiation	RadAvg	0.07	0.51	1.97	3.76	4.78	5.41	5.09	3.46	1.85	0.62	0.12	0.01	
	RadStd	0.01	0.09	0.23	0.29	0.40	0.51	0.54	0.35	0.18	0.11	0.02	0.00	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (5 neighbors)		+0.63	+2.50	+2.37	N/A	N/A	N/A	-158	-193	N/A	N/A	

Nomenclature: See separate page